Note for Integrating Factors:

We can take the integrating factor to be any non-zero constant multiple of an integrating factor. For example, $e^{\int \frac{1}{x} dx} = e^{\ln|x|} = |x|$

On any interval not containing $x = 0$, $|x|$ differs from x only by a constant factor (namely ± 1), so we may take the integrating factor to be x. So throughout these worked solutions, the modulus signs will be dropped when multiplying the differential equation through by the integrating factor.

1) a)

The 2nd order differential equation does not have constant coefficients, as $\frac{3}{x}$ is the coefficient of $\frac{dy}{dx}$

1) b)

$u = \frac{dy}{dx}$ and $\frac{du}{dx} = \frac{d^2y}{dx^2}$

$\frac{d^2y}{dx^2} + \frac{3}{x}\frac{dy}{dx} = x^2$

$\Rightarrow \frac{du}{dx} + \frac{3}{x}u = x^2$

1) c)

$\frac{du}{dx} + \frac{3}{x}u = x^2$

Integrating Factor

$= e^{\int \frac{3}{x} dx}$

$= e^{3\ln|x|}$

$= e^{\ln|x^3|}$

$= |x^3|$

Let Integrating Factor $= x^3$

$x^3\frac{du}{dx} + \frac{3}{x}u \times x^3 = x^2 \times x^3$

$\Rightarrow x^3\frac{du}{dx} + 3x^2u = x^5$

$\Rightarrow \frac{d}{dx}(x^3u) = x^5$

$\Rightarrow x^3u = \frac{1}{6}x^6 + c$

$\Rightarrow u = \frac{1}{6}x^3 + cx^{-3}$

1) d)

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{6}x^3 + cx^{-3}$

$\Rightarrow y = \frac{1}{24}x^4 - \frac{1}{2}cx^{-2} + k$

$\Rightarrow y = \frac{1}{24}x^4 + Ax^{-2} + k$

Where $A = -\frac{1}{2}c$

2) a)

The 2nd order differential equation does not have constant coefficients, as x is the coefficient of $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$

2) b)

$u = \frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}^2y}{\mathrm{d}x^2}$

$x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{1}{2}\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^{\frac{3}{2}}$

$\Rightarrow x\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{1}{2}u = 3x^{\frac{3}{2}}$

2) c)

$x\frac{\mathrm{d}u}{\mathrm{d}x}+\frac{1}{2}u=3x^{\frac{3}{2}}$

First, rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x}+P(x)u=Q(x)$:

$x\frac{\mathrm{d}u}{\mathrm{d}x}+\frac{1}{2}u=3x^{\frac{3}{2}}$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x}+\frac{1}{2x}u=3x^{\frac{1}{2}}$

Integrating Factor

$=\mathrm{e}^{\int\frac{1}{2x}\mathrm{d}x}$

$=\mathrm{e}^{\frac{1}{2}\ln|x|}$

$=\mathrm{e}^{\ln\left|x^{\frac{1}{2}}\right|}$

$=\left|x^{\frac{1}{2}}\right|$

Let Integrating Factor $=x^{\frac{1}{2}}$

$x^{\frac{1}{2}}\frac{\mathrm{d}u}{\mathrm{d}x}+\frac{1}{2x}u\times x^{\frac{1}{2}}=3x^{\frac{1}{2}}\times x^{\frac{1}{2}}$

$\Rightarrow x^{\frac{1}{2}}\frac{\mathrm{d}u}{\mathrm{d}x}+\frac{1}{2}x^{-\frac{1}{2}}u=3x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(x^{\frac{1}{2}}u\right)=3x$

$\Rightarrow x^{\frac{1}{2}}u=\frac{3}{2}x^{2}+c$

$\Rightarrow u=\frac{3}{2}x^{\frac{3}{2}}+cx^{-\frac{1}{2}}$

2) d)

$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{3}{2}x^{\frac{3}{2}}+cx^{-\frac{1}{2}}$

$\Rightarrow y=\frac{3}{5}x^{\frac{5}{2}}+2cx^{\frac{1}{2}}+k$

$\Rightarrow y=\frac{3}{5}x^{\frac{5}{2}}+Ax^{\frac{1}{2}}+k$

Where $A=2c$

3)

$u = \frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}^2y}{\mathrm{d}x^2}$

$(2x+1)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 3\frac{\mathrm{d}y}{\mathrm{d}x} = 4x+2$

$\Rightarrow (2x+1)\frac{\mathrm{d}u}{\mathrm{d}x} + 3u = 4x+2$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + P(x)u = Q(x)$:

$(2x+1)\frac{\mathrm{d}u}{\mathrm{d}x} + 3u = 2(2x+1)$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} + \frac{3}{2x+1}u = 2$

Integrating Factor

$= \mathrm{e}^{\int \frac{3}{2x+1}\mathrm{d}x}$

$= \mathrm{e}^{\frac{3}{2}\ln|2x+1|}$

$= \mathrm{e}^{\ln\left|(2x+1)^{\frac{3}{2}}\right|}$

$= \left|(2x+1)^{\frac{3}{2}}\right|$

Let Integrating Factor $= (2x+1)^{\frac{3}{2}}$

$(2x+1)^{\frac{3}{2}}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{3}{2x+1}u \times (2x+1)^{\frac{3}{2}} = 2 \times (2x+1)^{\frac{3}{2}}$

$\Rightarrow (2x+1)^{\frac{3}{2}}\frac{\mathrm{d}u}{\mathrm{d}x} + 3(2x+1)^{\frac{1}{2}}u = 2(2x+1)^{\frac{3}{2}}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left((2x+1)^{\frac{3}{2}}u\right) = 2(2x+1)^{\frac{3}{2}}$

$\Rightarrow (2x+1)^{\frac{3}{2}}u = \frac{2}{5}(2x+1)^{\frac{5}{2}} + c$

$\Rightarrow u = \frac{2}{5}(2x+1) + c(2x+1)^{-\frac{3}{2}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{5}(2x+1) + c(2x+1)^{-\frac{3}{2}}$

$\Rightarrow y = \frac{1}{10}(2x+1)^2 - c(2x+1)^{-\frac{1}{2}} + k$

4)

$u = \frac{dy}{dx}$ and $\frac{du}{dx} = \frac{d^2y}{dx^2}$

$(x^2 + 4)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 4x$

$\Rightarrow (x^2 + 4)\frac{du}{dx} + 2xu = 4x$

Rearrange into the form $\frac{du}{dx} + P(x)u = Q(x)$:

$(x^2 + 4)\frac{du}{dx} + 2xu = 4x$

$\Rightarrow \frac{du}{dx} + \frac{2x}{x^2+4}u = \frac{4x}{x^2+4}$

Integrating Factor

$= e^{\int \frac{2x}{x^2+4} dx}$

$= e^{\ln|x^2+4|}$

$= e^{\ln|x^2+4|}$

$= |x^2 + 4|$

Let Integrating Factor $= x^2 + 4$

$(x^2 + 4)\frac{du}{dx} + \frac{2x}{x^2+4}u \times (x^2 + 4) = \frac{4x}{x^2+4} \times (x^2 + 4)$

$\Rightarrow (x^2 + 4)\frac{du}{dx} + 2xu = 4x$

$\Rightarrow \frac{d}{dx}\left((x^2 + 4)u\right) = 4x$

$\Rightarrow (x^2 + 4)u = 2x^2 + c$

$\Rightarrow u = \frac{2x^2+c}{x^2+4}$

$\Rightarrow u = \frac{2(x^2+4)-8+c}{x^2+4}$

$\Rightarrow u = 2 + \frac{c-8}{x^2+4}$

$\Rightarrow \frac{dy}{dx} = 2 + \frac{c-8}{4\left(1+\frac{1}{4}x^2\right)}$

$\Rightarrow \frac{dy}{dx} = 2 + \frac{c-8}{4} \times \frac{1}{1+\left(\frac{1}{2}x\right)^2}$

$\Rightarrow y = 2x + \frac{c-8}{4} \times \frac{1}{\frac{1}{2}} \times \arctan\frac{1}{2}x + k$

$\Rightarrow y = 2x + A\arctan\frac{1}{2}x + k$

Where $A = \frac{c-8}{2}$

Substitute $y = 1$ and $x = 0$:

$1 = 2 \times 0 + A\arctan 0 + k \Rightarrow k = 1$

$\therefore y = 2x + A\arctan\frac{1}{2}x + 1$

$\frac{dy}{dx} = 2 + \frac{1}{2}A \times \frac{1}{1+\left(\frac{1}{2}x\right)^2}$

Substitute $\frac{dy}{dx} = 0$ and $x = 0$:

$0 = 2 + \frac{1}{2}A \times \frac{1}{1+\left(\frac{1}{2}\times 0\right)^2} \Rightarrow A = -4$

Particular Solution:

$y = 2x - 4\arctan\frac{1}{2}x + 1$

5) a)

$\frac{d}{dx}\left(\frac{x}{2}\tan x\right)$

$= \frac{x}{2} \times \sec^2 x + \frac{1}{2} \times \tan x$

$= \frac{1}{2}x\sec^2 x + \frac{1}{2}\tan x$

5) b)

$u = \frac{dy}{dx}$ and $\frac{du}{dx} = \frac{d^2y}{dx^2}$

$\cos x\frac{d^2y}{dx^2} - 2\sin x\frac{dy}{dx} = \cos x$

$\Rightarrow \cos x\frac{du}{dx} - 2u\sin x = \cos x$

Rearrange into the form $\frac{du}{dx} + P(x)u = Q(x)$:

$\cos x\frac{du}{dx} - 2u\sin x = \cos x$

$\Rightarrow \frac{du}{dx} - 2\frac{\sin x}{\cos x}u = 1$

Integrating Factor

$= e^{\int -2\frac{\sin x}{\cos x}dx}$

$= e^{2\ln|\cos x|}$

$= e^{\ln|\cos^2 x|}$

$= |\cos^2 x|$

Let Integrating Factor $= \cos^2 x$

$\cos^2 x\frac{du}{dx} - 2\frac{\sin x}{\cos x}u \times \cos^2 x = 1 \times \cos^2 x$

$\Rightarrow \cos^2 x\frac{du}{dx} - 2\sin x\cos x\, u = \cos^2 x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(u\cos^2 x) = \cos^2 x$

$\cos 2A \equiv 2\cos^2 A - 1 \Rightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$

$\therefore \frac{\mathrm{d}}{\mathrm{d}x}(u\cos^2 x) = \frac{1}{2}\cos 2x + \frac{1}{2}$

$\Rightarrow u\cos^2 x = \frac{1}{4}\sin 2x + \frac{1}{2}x + c$

$\Rightarrow u = \frac{\frac{1}{4}\sin 2x + \frac{1}{2}x + c}{\cos^2 x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{1}{2}\sin x\cos x + \frac{1}{2}x + c}{\cos^2 x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{1}{2}\sin x\cos x}{\cos^2 x} + \frac{\frac{1}{2}x}{\cos^2 x} + \frac{c}{\cos^2 x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}\tan x + \frac{1}{2}x\sec^2 x + c\sec^2 x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x}{2}\tan x\right) + c\sec^2 x$

$\Rightarrow y = \frac{1}{2}x\tan x + c\tan x + k$

Substitute $y = 2$ and $x = 0$:

$2 = \frac{1}{2}\times 0\times\tan 0 + c\tan 0 + k \Rightarrow k = 2$

$\therefore y = \frac{1}{2}x\tan x + c\tan x + 2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}x\sec^2 x + \frac{1}{2}\tan x + c\sec^2 x$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 3$ and $x = 0$:

$3 = \frac{1}{2}\times 0\times\sec^2 0 + \frac{1}{2}\tan 0 + c\sec^2 0 \Rightarrow c = 3$

Particular Solution:

$y = \frac{1}{2}x\tan x + 3\tan x + 2$

6)

$u = \frac{dy}{dx}$ and $\frac{du}{dx} = \frac{d^2y}{dx^2}$

$(x+2)(x+3)\frac{d^2y}{dx^2} + \frac{dy}{dx} = (x+3)^2$

$\Rightarrow (x+2)(x+3)\frac{du}{dx} + u = (x+3)^2$

Rearrange into the form $\frac{du}{dx} + P(x)u = Q(x)$:

$(x+2)(x+3)\frac{du}{dx} + u = (x+3)^2$

$\Rightarrow \frac{du}{dx} + \frac{u}{(x+2)(x+3)} = \frac{x+3}{x+2}$

Use Partial Fractions:

$\frac{1}{(x+2)(x+3)} \equiv \frac{A}{x+2} + \frac{B}{x+3}$

$\Rightarrow 1 \equiv A(x+3) + B(x+2)$

Let $x = -3$: $1 = -B \Rightarrow B = -1$

Let $x = -2$: $1 = A \Rightarrow A = 1$

$\therefore \frac{du}{dx} + \left(\frac{1}{x+2} - \frac{1}{x+3}\right)u = \frac{x+3}{x+2}$

Integrating Factor

$= e^{\int\left(\frac{1}{x+2} - \frac{1}{x+3}\right)dx}$

$= e^{\ln|x+2| - \ln|x+3|}$

$= e^{\ln\left|\frac{x+2}{x+3}\right|}$

$= \left|\frac{x+2}{x+3}\right|$

Let Integrating Factor $= \frac{x+2}{x+3}$

$\frac{x+2}{x+3}\frac{du}{dx} + \frac{u}{(x+2)(x+3)} \times \frac{x+2}{x+3} = \frac{x+3}{x+2} \times \frac{x+2}{x+3}$

$\Rightarrow \frac{x+2}{x+3}\frac{du}{dx} + \frac{u}{(x+3)^2} = 1$

$\Rightarrow \frac{d}{dx}\left(\left(\frac{x+2}{x+3}\right)u\right) = 1$

$\Rightarrow \left(\frac{x+2}{x+3}\right)u = x + c$

$\Rightarrow u = \frac{(x+3)(x+c)}{x+2}$

$\Rightarrow \frac{dy}{dx} = \frac{x^2+(3+c)x+3c}{x+2}$

$\frac{x^2+(3+c)x+3c}{x+2} \equiv \frac{x(x+2)+(x+2)+c(x+2)+c-2}{x+2} \equiv x + 1 + c + \frac{c-2}{x+2}$

Or using polynomial division:

	x	$1+c$	
x	x^2	$(1+c)x$	$c-2$
$+2$	$2x$	$2+2c$	

$\therefore \frac{x^2+(3+c)x+3c}{x+2} \equiv x+1+c+\frac{c-2}{x+2}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = x+1+c+\frac{c-2}{x+2}$

$\Rightarrow y = \frac{1}{2}x^2+(1+c)x+(c-2)\ln|x+2|+k$

Substitute $y=1$ and $x=0$:

$1 = \frac{1}{2}\times 0^2+(1+c)\times 0+(c-2)\ln|0+2|+k$

$\Rightarrow 1 = (c-2)\ln 2+k$

$\frac{\mathrm{d}y}{\mathrm{d}x} = x+1+c+\frac{c-2}{x+2}$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x}=3$ and $x=0$:

$3 = 0+1+c+\frac{c-2}{0+2}$

$\Rightarrow 2 = c+\frac{c-2}{2}$

$\Rightarrow 4 = 2c+c-2$

$\Rightarrow 3c = 6$

$\Rightarrow c = 2$

$\therefore 1 = (2-2)\ln 2+k \Rightarrow k=1$

Particular Solution:

$y = \frac{1}{2}x^2+3x+1$

7) a)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^4\frac{\mathrm{d}y}{\mathrm{d}x}\right) = x^4\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+4x^3\frac{\mathrm{d}y}{\mathrm{d}x}$

7) b)

$u = x^4 \frac{dy}{dx}$

$\Rightarrow \frac{du}{dx} = x^4 \frac{d^2y}{dx^2} + 4x^3 \frac{dy}{dx}$

So:

$x^4(x^2+1)\frac{d^2y}{dx^2} + 4x^3(x^2+1)\frac{dy}{dx} - 2x^5\frac{dy}{dx} = 2x(x^2+1)^2$

$\Rightarrow (x^2+1)\frac{du}{dx} - 2x^5\frac{dy}{dx} = 2x(x^2+1)^2$

As $u = x^4\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{u}{x^4}$

$\Rightarrow (x^2+1)\frac{du}{dx} - 2x^5 \times \frac{u}{x^4} = 2x(x^2+1)^2$

$\Rightarrow (x^2+1)\frac{du}{dx} - 2xu = 2x(x^2+1)^2$

Rearrange into the form $\frac{du}{dx} + P(x)u = Q(x)$:

$(x^2+1)\frac{du}{dx} - 2xu = 2x(x^2+1)^2$

$\Rightarrow \frac{du}{dx} - \frac{2x}{x^2+1}u = 2x(x^2+1)$

Integrating Factor

$= e^{\int -\frac{2x}{x^2+1}dx}$

$= e^{-\ln|x^2+1|}$

$= e^{\ln|(x^2+1)^{-1}|}$

$= \left|\frac{1}{x^2+1}\right|$

Let Integrating Factor $= \frac{1}{x^2+1}$

$\frac{1}{x^2+1}\frac{du}{dx} - \frac{2x}{x^2+1}u \times \frac{1}{x^2+1} = 2x(x^2+1) \times \frac{1}{x^2+1}$

$\Rightarrow \frac{1}{x^2+1}\frac{du}{dx} - \frac{2x}{(x^2+1)^2}u = 2x$

$\Rightarrow \frac{d}{dx}\left(\left(\frac{1}{x^2+1}\right)u\right) = 2x$

$\Rightarrow \left(\frac{1}{x^2+1}\right)u = x^2 + c$

$\Rightarrow u = (x^2+1)(x^2+c)$

$\Rightarrow x^4\frac{dy}{dx} = x^4 + (1+c)x^2 + c$

$\Rightarrow \frac{dy}{dx} = 1 + \frac{1+c}{x^2} + \frac{c}{x^4}$

$\Rightarrow y = x - \frac{1+c}{x} - \frac{c}{3x^3} + k$

8) a)

$u = \frac{dy}{dx} - 3x$

$\Rightarrow \frac{du}{dx} = \frac{d^2y}{dx^2} - 3$

So $\frac{dy}{dx} = u + 3x$ and $\frac{d^2y}{dx^2} = \frac{du}{dx} + 3$

Then:

$(1 + x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = 3 - 3x^2$

$\Rightarrow (1 + x^2)\left(\frac{du}{dx} + 3\right) - 2x(u + 3x) = 3 - 3x^2$

$\Rightarrow (1 + x^2)\frac{du}{dx} + 3(1 + x^2) - 2xu - 6x^2 = 3 - 3x^2$

$\Rightarrow (1 + x^2)\frac{du}{dx} + 3 + 3x^2 - 2xu = 3 + 3x^2$

$\Rightarrow (1 + x^2)\frac{du}{dx} - 2xu = 0$

$\Rightarrow (1 + x^2)\frac{du}{dx} = 2xu$

8) b)

Use Separation of Variables:

$\int \frac{1}{u} du = \int \frac{2x}{1+x^2} dx$

$\Rightarrow \ln|u| = \ln|1 + x^2| + c$

$\Rightarrow u = \pm e^{\ln|1+x^2|+c}$

$\Rightarrow u = A(1 + x^2)$

Now $u = \frac{dy}{dx} - 3x$ so:

$\frac{dy}{dx} - 3x = A + Ax^2$

Substitute $\frac{dy}{dx} = 2$ and $x = 0$:

$2 - 3 \times 0 = A + A \times 0^2 \Rightarrow A = 2$

$\therefore \frac{dy}{dx} - 3x = 2 + 2x^2$

$\Rightarrow \frac{dy}{dx} = 2x^2 + 3x + 2$

$\Rightarrow y = \frac{2}{3}x^3 + \frac{3}{2}x^2 + 2x + k$

Substitute $y = 1$ and $x = 0$:

$1 = \frac{2}{3} \times 0^3 + \frac{3}{2} \times 0^2 + 2 \times 0 + k \Rightarrow k = 1$

$\therefore y = \frac{2}{3}x^3 + \frac{3}{2}x^2 + 2x + 1$

9) a)

$u = \frac{dy}{dx} + 3y$

$\Rightarrow \frac{du}{dx} = \frac{d^2y}{dx^2} + 3\frac{dy}{dx}$

So $\frac{dy}{dx} = u - 3y$ and $\frac{d^2y}{dx^2} = \frac{du}{dx} - 3\frac{dy}{dx} = \frac{du}{dx} - 3(u - 3y) = \frac{du}{dx} - 3u + 9y$

Then:

$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = e^{-3x}$

$\Rightarrow \left(\frac{du}{dx} - 3u + 9y\right) + 6(u - 3y) + 9y = e^{-3x}$

$\Rightarrow \frac{du}{dx} - 3u + 9y + 6u - 18y + 9y = e^{-3x}$

$\Rightarrow \frac{du}{dx} + 3u = e^{-3x}$

9) b)

$\frac{du}{dx} + 3u = e^{-3x}$

Integrating Factor

$= e^{\int 3dx}$

$= e^{3x}$

$e^{3x}\frac{du}{dx} + 3e^{3x}u = e^{-3x} \times e^{3x}$

$\Rightarrow e^{3x}\frac{du}{dx} + 3e^{3x}u = 1$

$\Rightarrow \frac{d}{dx}(e^{3x}u) = 1$

$\Rightarrow e^{3x}u = x + c$

$\Rightarrow u = (x + c)e^{-3x}$

9) c)

Now $u = \frac{dy}{dx} + 3y$ so:

$\frac{dy}{dx} + 3y = (x + c)e^{-3x}$

Integrating Factor

$= e^{\int 3dx}$

$= e^{3x}$

$e^{3x}\frac{dy}{dx} + 3e^{3x}y = (x + c)e^{-3x} \times e^{3x}$

$\Rightarrow e^{3x}\frac{dy}{dx} + 3e^{3x}y = x + c$

$\Rightarrow \frac{d}{dx}(e^{3x}y) = x + c$

$\Rightarrow e^{3x}y = \frac{1}{2}x^2 + cx + k$

$\Rightarrow y = \left(\frac{1}{2}x^2 + cx + k\right)e^{-3x}$

10) a)

$u = \ln x$

$\Rightarrow \frac{du}{dx} = \frac{1}{x}$

By the Chain Rule, we can write:

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{1}{x}$

$\Rightarrow \frac{dy}{dx} = \frac{1}{x}\frac{dy}{du}$

10) b)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}u}\right)$

Use the Product Rule and note that:

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}y}{\mathrm{d}u}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(y'(u)\right) = \frac{\mathrm{d}u}{\mathrm{d}x}y''(u) = \frac{\mathrm{d}^2y}{\mathrm{d}u^2}\frac{\mathrm{d}u}{\mathrm{d}x}$

Then:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}u}\right)$

$= -\frac{1}{x^2}\frac{\mathrm{d}y}{\mathrm{d}u} + \frac{1}{x}\frac{\mathrm{d}^2y}{\mathrm{d}u^2}\frac{\mathrm{d}u}{\mathrm{d}x}$

Recall that $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ so:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{x^2}\frac{\mathrm{d}y}{\mathrm{d}u} + \frac{1}{x}\frac{\mathrm{d}^2y}{\mathrm{d}u^2} \times \frac{1}{x}$

$= -\frac{1}{x^2}\frac{\mathrm{d}y}{\mathrm{d}u} + \frac{1}{x^2}\frac{\mathrm{d}^2y}{\mathrm{d}u^2}$

$= \frac{1}{x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}u^2} - \frac{\mathrm{d}y}{\mathrm{d}u}\right)$ as required

11)

$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x} - 4y = 0$

$\Rightarrow x^2 \times \frac{1}{x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}u^2} - \frac{\mathrm{d}y}{\mathrm{d}u}\right) + x \times \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}u} - 4y = 0$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}u^2} - \frac{\mathrm{d}y}{\mathrm{d}u} + \frac{\mathrm{d}y}{\mathrm{d}u} - 4y = 0$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}u^2} - 4y = 0$

Auxiliary Equation:

$m^2 - 4 = 0$

$\Rightarrow (m-2)(m+2) = 0$

$\Rightarrow m = 2$ or $m = -2$

$\therefore y = A\mathrm{e}^{2u} + B\mathrm{e}^{-2u}$

Now, $u = \mathrm{e}^u$ so:

$y = A(\mathrm{e}^u)^2 + B(\mathrm{e}^u)^{-2}$

$\Rightarrow y = Ax^2 + Bx^{-2}$

12)

$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - x\frac{\mathrm{d}y}{\mathrm{d}x} - 8y = 4$

$\Rightarrow x^2 \times \frac{1}{x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}u^2} - \frac{\mathrm{d}y}{\mathrm{d}u}\right) - x \times \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}u} - 8y = 4$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}u^2} - \frac{\mathrm{d}y}{\mathrm{d}u} - \frac{\mathrm{d}y}{\mathrm{d}u} - 8y = 4$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}u^2} - 2\frac{\mathrm{d}y}{\mathrm{d}u} - 8y = 4$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}u^2} - 2\frac{\mathrm{d}y}{\mathrm{d}u} - 8y = 0$

Auxiliary Equation:

$m^2 - 2m - 8 = 0$

$\Rightarrow (m-4)(m+2) = 0$

$\Rightarrow m = 4$ or $m = -2$

Complementary Function:

$y = A\mathrm{e}^{4u} + B\mathrm{e}^{-2u}$

Try this Particular Integral:

$y = k$

$\frac{\mathrm{d}y}{\mathrm{d}u} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}u^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}u^2} - 2\frac{\mathrm{d}y}{\mathrm{d}u} - 8y = 4$

$\Rightarrow 0 - 2(0) - 8(k) = 4$

$\Rightarrow -8k = 4$

$\Rightarrow k = -\frac{1}{2}$

Particular Integral:

$y = -\frac{1}{2}$

General Solution:

$y = A\mathrm{e}^{4u} + B\mathrm{e}^{-2u} - \frac{1}{2}$

Now, $x = \mathrm{e}^u$ so:

$y = A(\mathrm{e}^u)^4 + B(\mathrm{e}^u)^{-2} - \frac{1}{2}$

$\Rightarrow y = Ax^4 + Bx^{-2} - \frac{1}{2}$

13)

$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+3x\frac{\mathrm{d}y}{\mathrm{d}x}-3y=5x^2$

$\Rightarrow x^2\times\frac{1}{x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}u^2}-\frac{\mathrm{d}y}{\mathrm{d}u}\right)+3x\times\frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}u}-3y=5(\mathrm{e}^u)^2$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}u^2}-\frac{\mathrm{d}y}{\mathrm{d}u}+3\frac{\mathrm{d}y}{\mathrm{d}u}-3y=5\mathrm{e}^{2u}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}u^2}+2\frac{\mathrm{d}y}{\mathrm{d}u}-3y=5\mathrm{e}^{2u}$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}u^2}+2\frac{\mathrm{d}y}{\mathrm{d}u}-3y=0$

Auxiliary Equation:

$m^2+2m-3=0$

$\Rightarrow (m+3)(m-1)=0$

$\Rightarrow m=-3$ or $m=1$

Complementary Function:

$y=A\mathrm{e}^{-3u}+B\mathrm{e}^u$

Try this Particular Integral:

$y=k\mathrm{e}^{2u}$

$\frac{\mathrm{d}y}{\mathrm{d}u}=2k\mathrm{e}^{2u}$

$\frac{\mathrm{d}^2y}{\mathrm{d}u^2}=4k\mathrm{e}^{2u}$

$\frac{\mathrm{d}^2y}{\mathrm{d}u^2}+2\frac{\mathrm{d}y}{\mathrm{d}u}-3y=5\mathrm{e}^{2u}$

$\Rightarrow (4k\mathrm{e}^{2u})+2(2k\mathrm{e}^{2u})-3(k\mathrm{e}^{2u})=5\mathrm{e}^{2u}$

$\Rightarrow (4+4-3)k\mathrm{e}^{2u}=5\mathrm{e}^{2u}$

$\Rightarrow 5k\mathrm{e}^{2u}=5\mathrm{e}^{2u}$

$\Rightarrow k=1$

Particular Integral:

$y=\mathrm{e}^{2u}$

General Solution:

$y=A\mathrm{e}^{-3u}+B\mathrm{e}^u+\mathrm{e}^{2u}$

Now, $x=\mathrm{e}^u$ so:

$y=A(\mathrm{e}^u)^{-3}+B\mathrm{e}^u+(\mathrm{e}^u)^2$

$\Rightarrow y=Ax^{-3}+Bx+x^2$

Substitute $y=2$ and $x=1$:

$2=A\times1^{-3}+B\times1+1^2$

$\Rightarrow \mathrm{A}+\mathrm{B}=1$ (1)

$\frac{\mathrm{d}y}{\mathrm{d}x} = -3Ax^{-4} + B + 2x$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$ and $x = 1$:

$1 = -3A \times 1^{-4} + B + 2 \times 1$

$\Rightarrow -3A + B = -1$ (2)

Solve (1) & (2)

$A = \frac{1}{2}$ and $B = \frac{1}{2}$

Particular Solution:

$y = \frac{1}{2}x^{-3} + \frac{1}{2}x + x^2$

14) a)

$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4x\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 12 + 10\cos(\ln x)$

$\Rightarrow x^2 \times \frac{1}{x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t}\right) - 4x \times \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}t} + 6y = 12 + 10\cos(\ln \mathrm{e}^t)$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t} - 4\frac{\mathrm{d}y}{\mathrm{d}t} + 6y = 12 + 10\cos(t)$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\frac{\mathrm{d}y}{\mathrm{d}t} + 6y = 12 + 10\cos(t)$

14) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\frac{\mathrm{d}y}{\mathrm{d}t} + 6y = 0$

Auxiliary Equation:

$m^2 - 5m + 6 = 0$

$\Rightarrow (m-2)(m-3) = 0$

$\Rightarrow m = 2$ or $m = 3$

Complementary Function:

$y = A\mathrm{e}^{2t} + B\mathrm{e}^{3t}$

Try this Particular Integral:

$y = k + p\cos t + q\sin t$

$\frac{\mathrm{d}y}{\mathrm{d}t} = -p\sin t + q\cos t$

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -p\cos t - q\sin t$

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\frac{\mathrm{d}y}{\mathrm{d}t} + 6y = 12 + 10\cos t$

$\Rightarrow (-p\cos t - q\sin t) - 5(-p\sin t + q\cos t) + 6(k + p\cos t + q\sin t) = 12 + 10\cos t$

$\Rightarrow (6k) + (-p - 5q + 6p)\cos t + (-q + 5p + 6q)\sin t = 12 + 10\cos t$

$\Rightarrow (6k) + (5p - 5q)\cos t + (5p + 5q)\sin t = 12 + 10\cos t$

Compare Coefficients:

$6k = 12 \Rightarrow k = 2$

$5p - 5q = 10$ (1)

$5p + 5q = 0$ (2)

Solve (1) & (2)

$p = 1$ and $q = -1$

Particular Integral:

$y = 2 + \cos t - \sin t$

General Solution:

$y = A\mathrm{e}^{2t} + B\mathrm{e}^{3t} + 2 + \cos t - \sin t$

Now, $x = \mathrm{e}^{tt} \Rightarrow t = \ln x$ so:

$y = A\mathrm{e}^{2\ln x} + B\mathrm{e}^{3\ln x} + 2 + \cos(\ln x) - \sin(\ln x)$

$\Rightarrow y = A\mathrm{e}^{\ln x^2} + B\mathrm{e}^{\ln x^3} + 2 + \cos(\ln x) - \sin(\ln x)$

$\Rightarrow y = Ax^2 + Bx^3 + 2 + \cos(\ln x) - \sin(\ln x)$

15) a)

$x = \mathrm{e}^{3t}$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = 3\mathrm{e}^{3t}$

$\Rightarrow \frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{3\mathrm{e}^{3t}} = \frac{1}{3x}$

By the Chain Rule, we can write:

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}t} \times \frac{\mathrm{d}t}{\mathrm{d}x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}t} \times \frac{1}{3x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3x}\frac{\mathrm{d}y}{\mathrm{d}t}$

15) b)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{3x}\frac{\mathrm{d}y}{\mathrm{d}t}\right)$

Use the Product Rule and note that:

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(y'(t)\right) = \frac{\mathrm{d}t}{\mathrm{d}x}y''(t) = \frac{\mathrm{d}^2y}{\mathrm{d}t^2}\frac{\mathrm{d}t}{\mathrm{d}x}$

Then:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{3x}\frac{\mathrm{d}y}{\mathrm{d}t}\right)$

$= -\frac{1}{3x^2}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{3x}\frac{\mathrm{d}^2y}{\mathrm{d}t^2}\frac{\mathrm{d}t}{\mathrm{d}x}$

Recall that $\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{3x}$ so:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{3x^2}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{3x}\frac{\mathrm{d}^2y}{\mathrm{d}t^2} \times \frac{1}{3x}$

$= -\frac{1}{3x^2}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{9x^2}\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

$= \frac{1}{9x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 3\frac{\mathrm{d}y}{\mathrm{d}t}\right)$ as required

16) a)

$9x^3\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2xy=x(\ln x)^2+20$

$\Rightarrow 9x^3\times\frac{1}{9x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2}-3\frac{\mathrm{d}y}{\mathrm{d}t}\right)+2xy=x(\ln x)^2+20$

$\Rightarrow x\left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2}-3\frac{\mathrm{d}y}{\mathrm{d}t}\right)+2xy=x(\ln x)^2+20$

$\Rightarrow \mathrm{e}^{3t}\left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2}-3\frac{\mathrm{d}y}{\mathrm{d}t}\right)+2\mathrm{e}^{3t}y=\mathrm{e}^{3t}(\ln(\mathrm{e}^{3t}))^2+20$

$\Rightarrow \mathrm{e}^{3t}\frac{\mathrm{d}^2y}{\mathrm{d}t^2}-3\mathrm{e}^{3t}\frac{\mathrm{d}y}{\mathrm{d}t}+2\mathrm{e}^{3t}y=\mathrm{e}^{3t}(3t)^2+20$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2}-3\frac{\mathrm{d}y}{\mathrm{d}t}+2y=9t^2+20\mathrm{e}^{-3t}$

16) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2}-3\frac{\mathrm{d}y}{\mathrm{d}t}+2y=0$

Auxiliary Equation:

$m^2-3m+2=0$

$\Rightarrow (m-1)(m-2)=0$

$\Rightarrow m=1$ or $m=2$

Complementary Function:

$y=A\mathrm{e}^t+B\mathrm{e}^{2t}$

Try this Particular Integral:

$y=pt^2+qt+r+k\mathrm{e}^{-3t}$

$\frac{\mathrm{d}y}{\mathrm{d}t}=2pt+q-3k\mathrm{e}^{-3t}$

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2}=2p+9k\mathrm{e}^{-3t}$

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2}-3\frac{\mathrm{d}y}{\mathrm{d}t}+2y=9t^2+20\mathrm{e}^{-3t}$

$\Rightarrow (2p+9k\mathrm{e}^{-3t})-3(2pt+q-3k\mathrm{e}^{-3t})+2(pt^2+qt+r+k\mathrm{e}^{-3t})=9t^2+20\mathrm{e}^{-3t}$

$\Rightarrow (2p)t^2+(-6p+2q)t+(2p-3q+2r)+(9+9+2)k\mathrm{e}^{-3t}=9t^2+20\mathrm{e}^{-3t}$

$\Rightarrow (2p)t^2+(-6p+2q)t+(2p-3q+2r)+20k\mathrm{e}^{-3t}=9t^2+20\mathrm{e}^{-3t}$

Compare Coefficients:

$2p=9\Rightarrow p=\frac{9}{2}$

$-6\times\frac{9}{2}+2q=0\Rightarrow q=\frac{27}{2}$

$2\times\frac{9}{2}-3\times\frac{27}{2}+2r=0\Rightarrow r=\frac{63}{4}$

$20k=20\Rightarrow k=1$

Particular Integral:

$y = \frac{9}{2}t^2 + \frac{27}{2}t + \frac{63}{.4} + \mathrm{e}^{-3t}$

General Solution:

$y = A\mathrm{e}^t + B\mathrm{e}^{2t} + \frac{9}{2}t^2 + \frac{27}{2}t + \frac{63}{.4} + \mathrm{e}^{-3t}$

Now, $x = \mathrm{e}^{3t} \Rightarrow t = \frac{1}{3}\ln x$ so:

$y = A\mathrm{e}^{\frac{1}{3}\ln x} + B\mathrm{e}^{\frac{2}{3}\ln x} + \frac{9}{2}\left(\frac{1}{3}\ln x\right)^2 + \frac{27}{2}\left(\frac{1}{3}\ln x\right) + \frac{63}{4} + \frac{1}{x}$

$\Rightarrow y = A\mathrm{e}^{\ln x^{\frac{1}{3}}} + B\mathrm{e}^{\ln x^{\frac{2}{3}}} + \frac{9}{2}\left(\frac{1}{3}\ln x\right)^2 + \frac{27}{2}\left(\frac{1}{3}\ln x\right) + \frac{63}{4} + \frac{1}{x}$

$\Rightarrow y = Ax^{\frac{1}{3}} + Bx^{\frac{2}{3}} + \frac{1}{2}(\ln x)^2 + \frac{9}{2}\ln x + \frac{63}{4} + \frac{1}{x}$

17) a)

$y = u\cos x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -u\sin x + \cos x\frac{\mathrm{d}u}{\mathrm{d}x}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -u\cos x - \sin x\frac{\mathrm{d}u}{\mathrm{d}x} + \cos x\frac{\mathrm{d}^2u}{\mathrm{d}x^2} - \sin x\frac{\mathrm{d}u}{\mathrm{d}x}$

$= -u\cos x - 2\sin x\frac{\mathrm{d}u}{\mathrm{d}x} + \cos x\frac{\mathrm{d}^2u}{\mathrm{d}x^2}$

So:

$\cos^2 x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \sin 2x\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = \sin 2x\cos^3 x$

$\Rightarrow \cos^2 x\left(-u\cos x - 2\sin x\frac{\mathrm{d}u}{\mathrm{d}x} + \cos x\frac{\mathrm{d}^2u}{\mathrm{d}x^2}\right) + 2\sin x\cos x\left(-u\sin x + \cos x\frac{\mathrm{d}u}{\mathrm{d}x}\right) + 2(u\cos x) = 2\sin x\cos x \times \cos^3 x$

$\Rightarrow -u\cos^3 x - 2\sin x\cos^2 x\frac{\mathrm{d}u}{\mathrm{d}x} + \cos^3 x\frac{\mathrm{d}^2u}{\mathrm{d}x^2} - 2u\sin^2 x\cos x + 2\sin x\cos^2 x\frac{\mathrm{d}u}{\mathrm{d}x} + 2u\cos x = 2\sin x\cos^4 x$

$\Rightarrow -u\cos^2 x + \cos^2 x\frac{\mathrm{d}^2u}{\mathrm{d}x^2} - 2u\sin^2 x + 2u = 2\sin x\cos^3 x$

$\Rightarrow \cos^2 x\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + (2 - \cos^2 x - 2\sin^2 x)u = 2\sin x\cos^3 x$

$\Rightarrow \cos^2 x\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + (2\sin^2 x + 2\cos^2 x - \cos^2 x - 2\sin^2 x)u = 2\sin x\cos^3 x$

$\Rightarrow \cos^2 x\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + (\cos^2 x)u = 2\sin x\cos^3 x$

$\Rightarrow \frac{\mathrm{d}^2u}{\mathrm{d}x^2} + u = 2\sin x\cos x$

$\Rightarrow \frac{\mathrm{d}^2u}{\mathrm{d}x^2} + u = \sin 2x$

17) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + u = 0$

Auxiliary Equation:

$m^2 + 1 = 0$

$\Rightarrow m^2 = -1$

$\Rightarrow m = \pm \mathrm{i}$

Complementary Function:

$u = A\cos x + B\sin x$

Try this Particular Integral:

$u = p\cos 2x + q\sin 2x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -2p\sin 2x + 2q\cos 2x$

$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} = -4p\cos 2x - 4q\sin 2x$

$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + u = \sin 2x$

$\Rightarrow (-4p\cos 2x - 4q\sin 2x) + (p\cos 2x + q\sin 2x) = \sin 2x$

$\Rightarrow (-4p + p)\cos 2x + (-4q + q)\sin 2x = \sin 2x$

$\Rightarrow (-3p)\cos 2x + (-3q)\sin 2x = \sin 2x$

Compare Coefficients:

$-3p = 0 \Rightarrow p = 0$

$-3q = 1 \Rightarrow q = -\frac{1}{3}$

Particular Integral:

$u = -\frac{1}{3}\sin 2x$

General Solution:

$u = A\cos x + B\sin x - \frac{1}{3}\sin 2x$

Now $y = u\cos x \Rightarrow u = \frac{y}{\cos x}$ so:

$\frac{y}{\cos x} = A\cos x + B\sin x - \frac{1}{3}\sin 2x$

$\Rightarrow y = A\cos^2 x + B\sin x\cos x - \frac{1}{3}\sin 2x\cos x$